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EXPERIMENT 7

AIM: Implement the embedding wallet (Metamask) and transaction using Solidity

Theory:

Blockchain is a decentralized and distributed digital ledger that securely records transactions across multiple systems. It ensures transparency, security, and immutability, meaning that once data is recorded, it cannot be altered.

Ethereum is a widely used blockchain platform that enables the execution of smart contracts. Smart contracts are self-executing programs written in Solidity that automatically carry out actions when predefined conditions are met, eliminating the need for intermediaries.

MetaMask is a cryptocurrency wallet and browser extension used to interact with blockchain networks. It securely stores user accounts and private keys, and allows users to send, receive, and sign transactions. It also serves as a bridge between Remix IDE and blockchain networks.

Test networks such as Sepolia and RSK Testnet are used for development and testing. They provide free tokens, enabling developers to deploy and test smart contracts without using real cryptocurrency, thereby reducing financial risk.

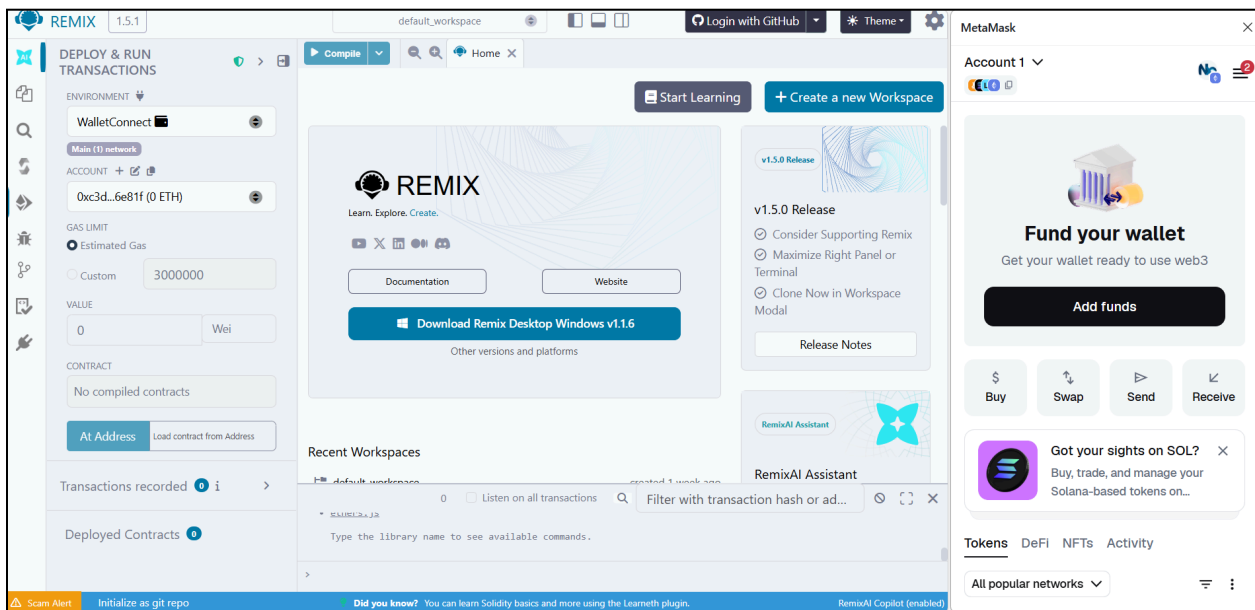
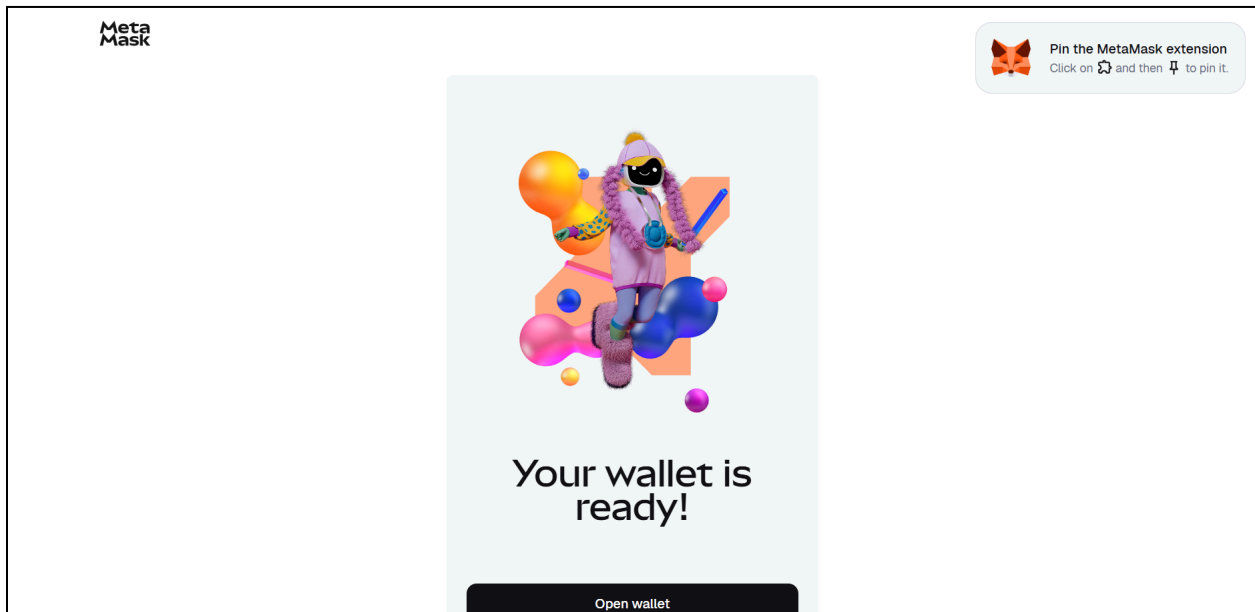
Remix IDE is a web-based development environment used to write, compile, deploy, and test smart contracts. It offers a user-friendly interface and integrates directly with MetaMask for deploying contracts on blockchain networks.

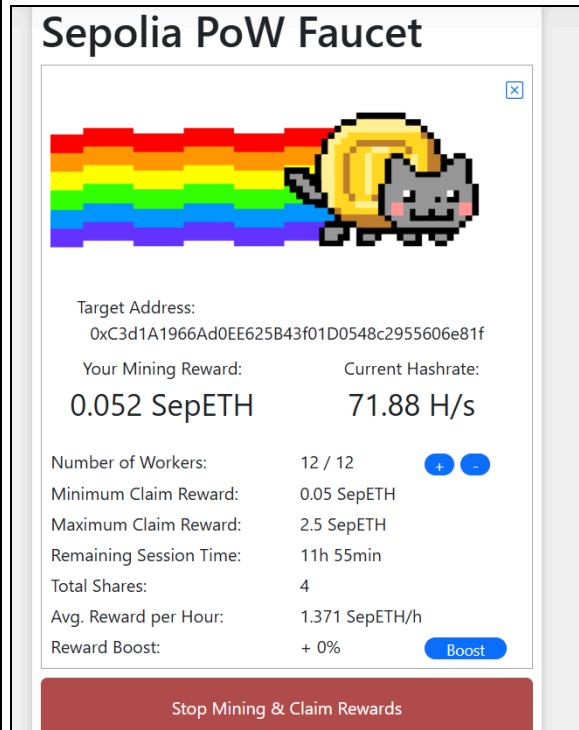
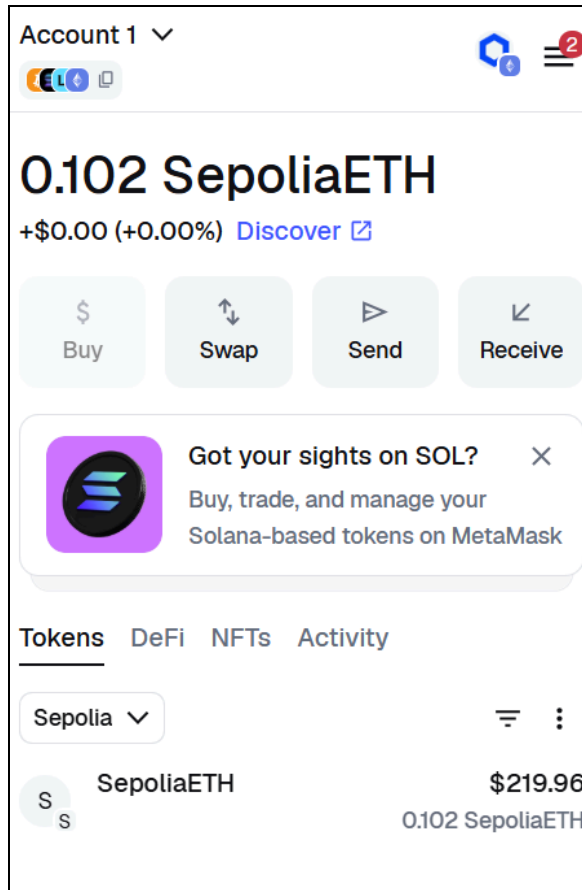
The smart contract deployment process involves writing the contract in Solidity, compiling it to eliminate errors, connecting MetaMask to Remix, and deploying the contract on a testnet by paying gas fees with test tokens.

A blockchain transaction refers to any interaction with the network, such as deploying a contract or executing its functions. Each transaction requires authentication through a digital signature and is permanently recorded on the blockchain. These transactions can be verified using blockchain explorers.

Step 1 —Install MetaMask

Install Metamask and create wallet





Step 2 — Deploy BugBountyToken First

1. In Remix, go to the **Deploy tab** (Ethereum icon on left)
2. Make sure Environment is "**Injected Provider - MetaMask**"
3. In the **Contract dropdown**, select "**BugBountyToken**"
4. Click **Deploy**
5. MetaMask will pop up → click **Confirm**
6. Wait a few seconds → you'll see it appear under "**Deployed Contracts**"
7. **Copy and save its address** (click the copy icon next to it)

REMITX1.5.1

default_workspace

Login with GitHubTheme

DEPLOY & RUN TRANSACTIONS

ACCOUNT0xc3d...6e81f (0 ETH)

GAS LIMIT

Estimated Gas

Custom3000000

VALUE

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CONTRACT

BugBountyToken - BugBountyToken.s

even version: osaka

Verify Contract on Explorers

Deploy & Verify

At AddressLoad contract from Address

Transactions recorded1i

Deployed Contracts0

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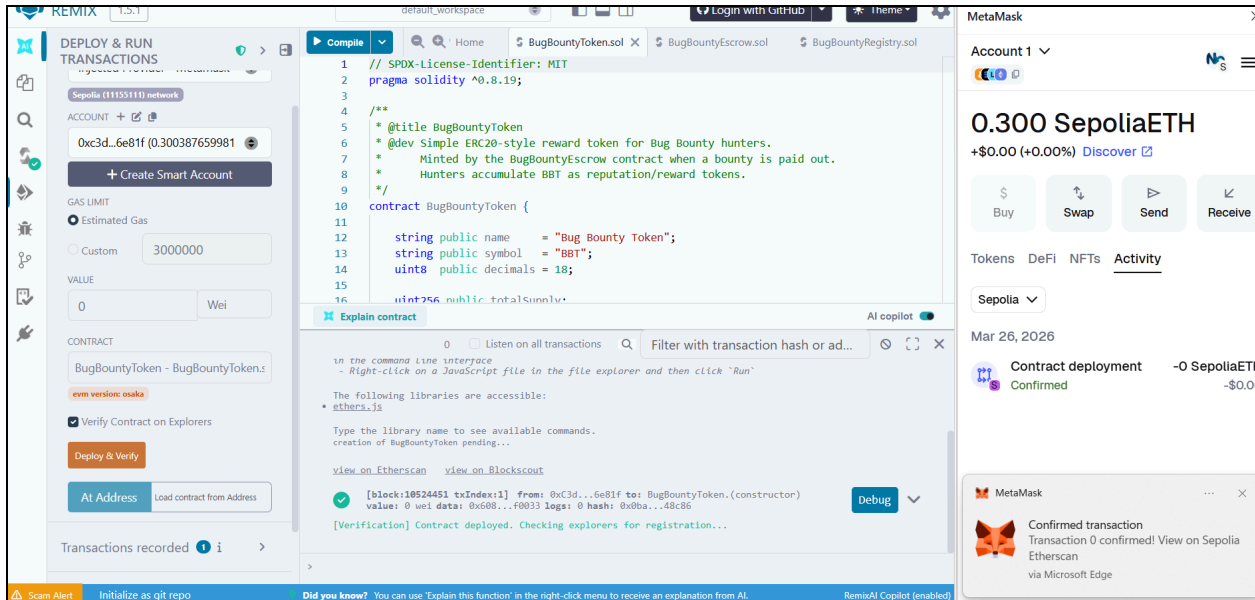
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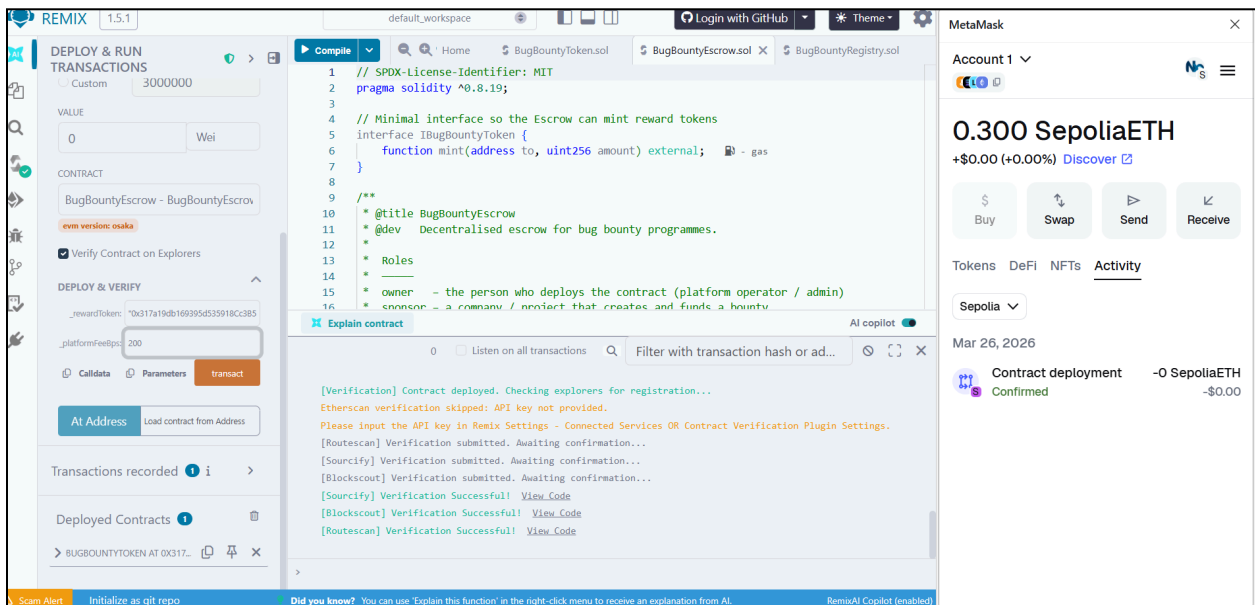
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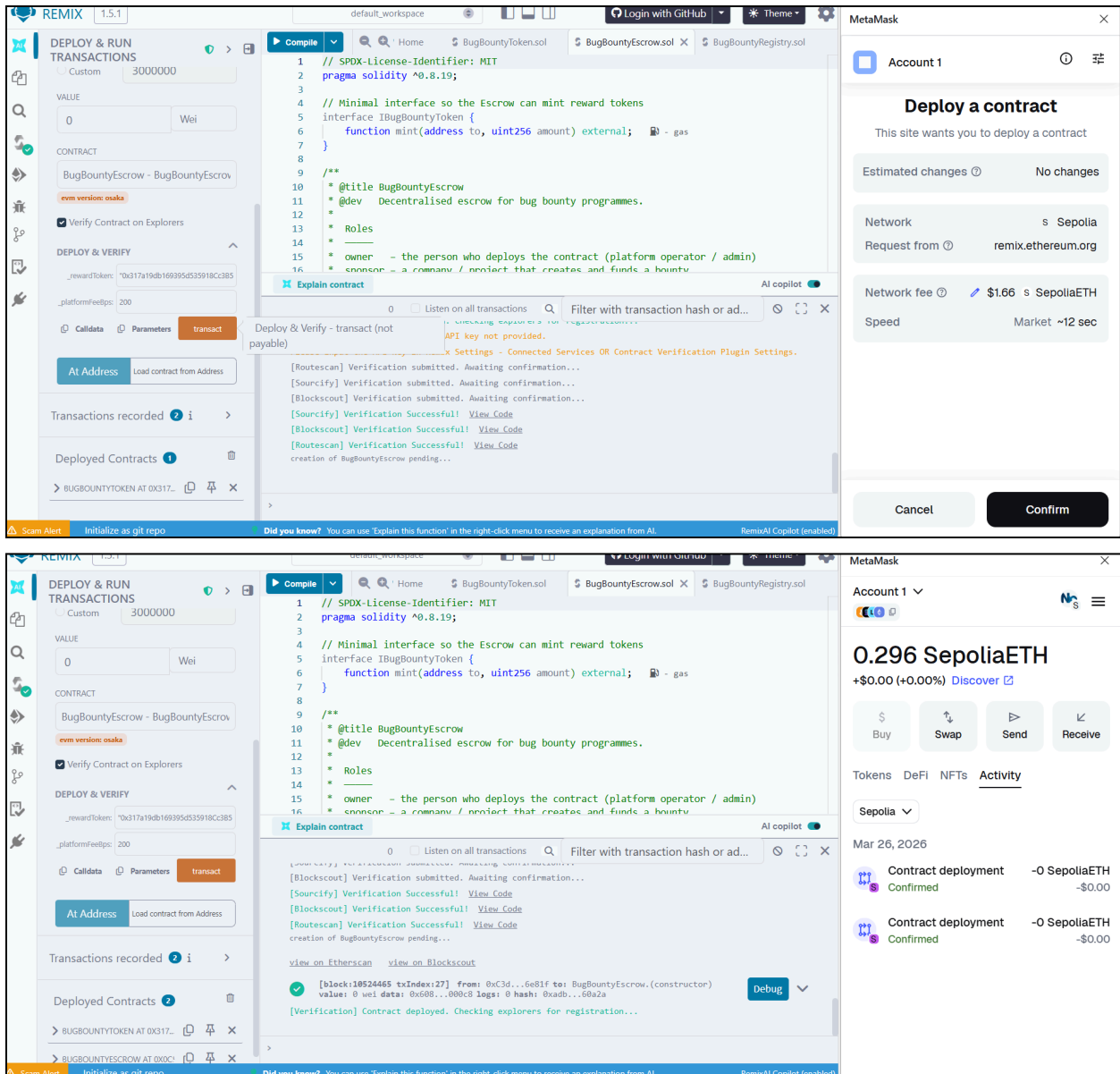
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Step 3 — Deploy BugBountyEscrow

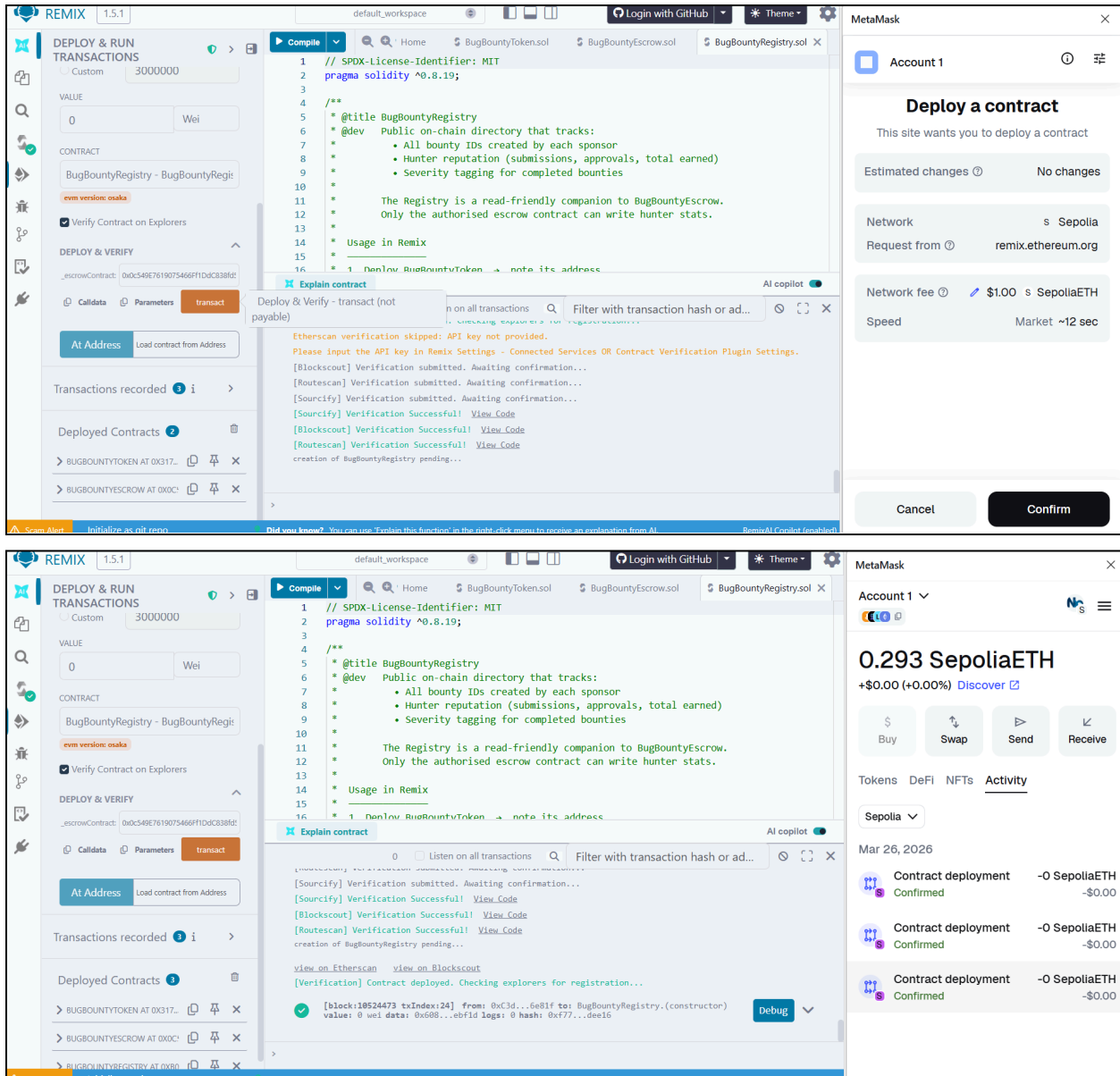
1. In the Contract dropdown, select "**BugBountyEscrow**"
2. You'll see two input fields before deploying:
 - `_rewardToken` → paste the **BugBountyToken** address from Step 1
 - `_platformFeeBps` → type **200** (means 2% fee)
3. Click **Deploy** → Confirm in MetaMask
4. **Copy and save its address**





Step 4 — Deploy BugBountyRegistry

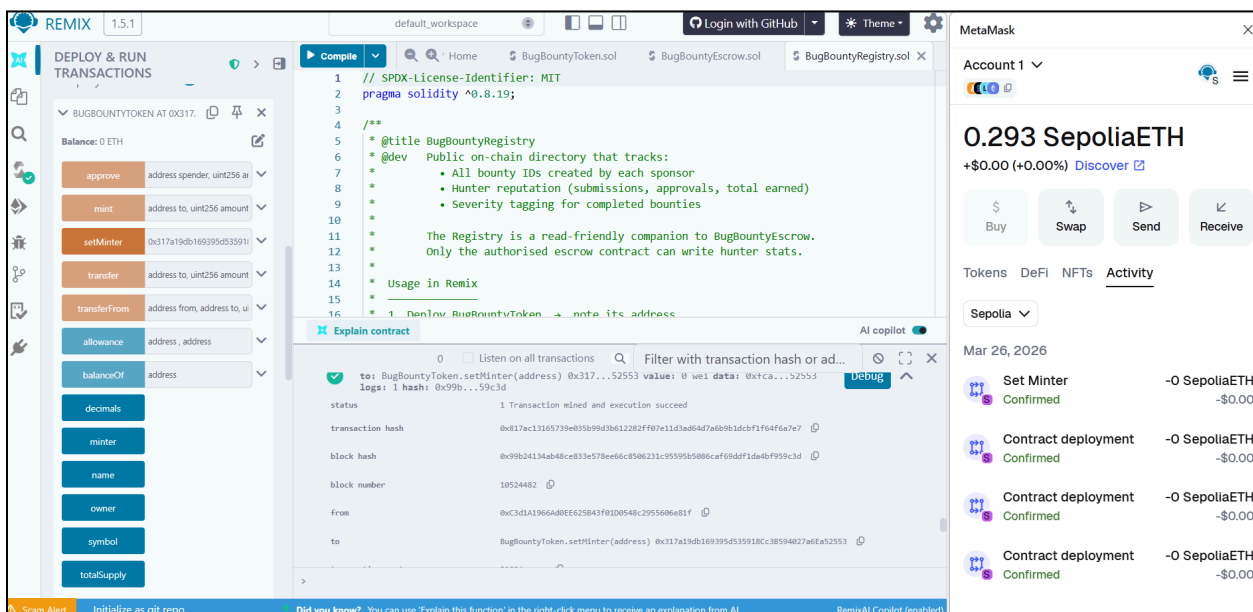
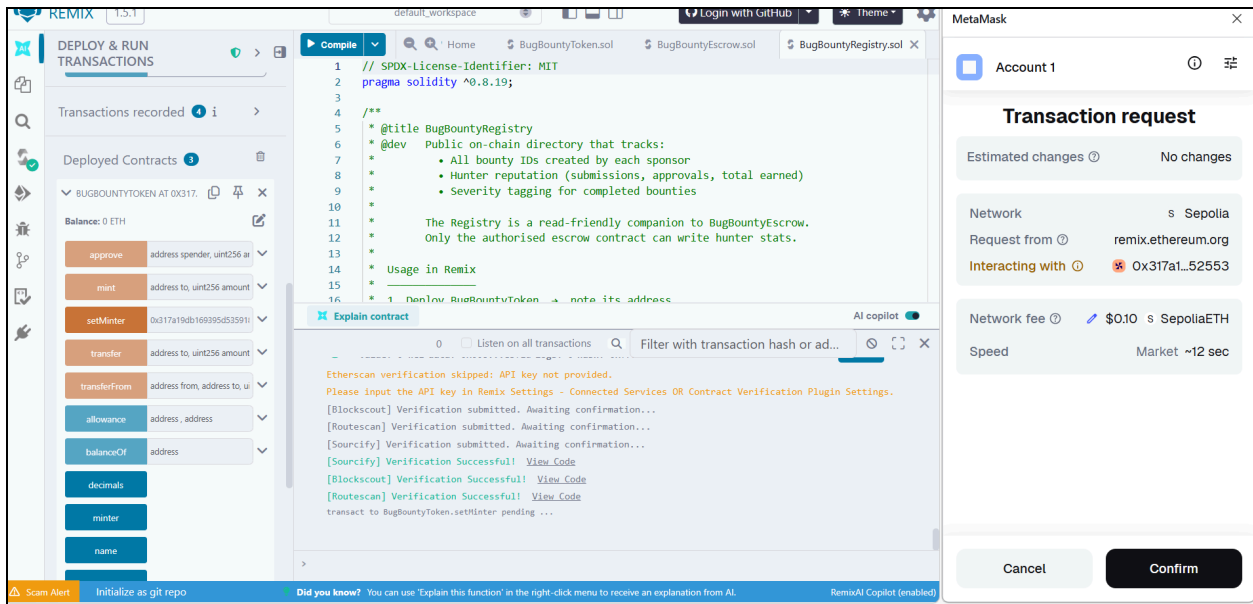
1. Select "**BugBountyRegistry**" in the dropdown
2. Input field:
 - `_escrowContract` → paste the **BugBountyEscrow** address from Step 2
3. Click **Deploy** → Confirm in MetaMask
4. **Copy and save its address**



Step 5 — Link the Contracts Together

On BugBountyToken:

1. Expand it under Deployed Contracts
2. Find **setMinter** function
3. Paste the **BugBountyEscrow** address → click it → Confirm in MetaMask



Conclusion

The experiment successfully demonstrated MetaMask integration with Remix IDE and execution of blockchain transactions. It provided hands-on experience with smart contracts, ensuring secure and transparent interactions on a testnet